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MILITARY SPECIFICATION

LOADER-TRANSPORTER, GUIDED MISSILE: XM501E3 (IMPROVED HAWK)

This specification is approved for use by the US Army Tank-Automotive -Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of a full-tracked, self-propelled, tactical missile loader for loading and unloading Hawk Guided Missiles on to and from Hawk Missile System Launchers and Pallets.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertdnent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1450

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SPECIFICATIONS
MILITARY

MIL-P-514	Plates; Identification, Transportation-Data and Blank.
MIL-H-5440	Hydraulic Systems, Aircraft Types I and II, Design, Installation, and Data Requirements for.
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance,
MIL-L-45779	Loader-Transporter, Guided Missile, XM501E2 and XM501E3, Processing for Storage and Shipment of.
MIL-E-55301	Electromagnetic Comparability.

STANDARDS
MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-193	Painting Procedures Tactical (Tracked and Wheeled).
MIL-STD-642	Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. .

DRAWINGS

10879601	- Alignment Checkout Fixture.
11675300	- Loader-Transporter, Guided Missile: XM501E3.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the acquisitioning activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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MISSILE PURCHASE DESCRIPTION

MPD-7005 - Limit Environmental Criteria for the Anti-aircraft Guided Missile System - Hawk.

(Application for copies should be addressed to the US Army Missile Command, AMSMI-IDS Redstone Arsenal, Alabama 35809.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article Unless otherwise specified (see 6.2) the contractor shall furnish guided missile loader-transporters under this specification which shall be subjected to first article inspection (see 4.4). The first article inspection vehicle, properly marked with identifying information shall be representative of the vehicles to be furnished to the Government in production quantities. Any changes or deviations of the production vehicles from the first article sample shall be subject to the approval of the contracting officer (see 4.4).

3.2 Materials.

3.201 Qualified products. With respect to all assemblies and component parts requiring product qualification in accordance with the pertinent specifications or drawings (see 303), and necessary in the construction of this vehicle, the contractor shall be responsible for using only those products which are listed on the Qualified Parts List(QPL) in the referenced specification or which have been approved for inclusion on such Lists. Manufacturer's inspection records shall specifically list such components~ name of suppliers, and number and date of the QPL from which the selection was made or date of approval letter when products are approved but not as yet listed on the QPL.

3.3 Design and construction. Vehicles, components, sub-assemblies, and assemblies shall be fabricated and assembled into a complete loader in accordance with drawing 11675300. All parts requiring identification shall be identified in accordance with MIL-STD-130, unless otherwise specified on the applicable drawings

3.3.1 Environmental. The loader shall perform satisfactorily when subjected to applicable environmental conditions specified in Purchase Description MPD-7005 (see 4.6.3.1).

3*3.2 Fuel system. The fuel system shall maintain a fuel supply to the engine when ascending and descending 60 percent grades in forward and reverse gear, and when the vehicle is being operated on 30 percent side slopes, in either direction (see 4.6.3.2).

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3.3.2.1 Fuel tank and lines. Provisions shall be made to assure the internal cleanliness of the fuel tank, lines, and connections prior to initial fueling of the vehicle. Fuel tank, lines, and connections shall be free from leakage after installation and after pre-test run (see 4.6.1.1).

3.3.3 Hydraulic system. The loader hydraulic system shall conform to type II, 3000 psi class of MIL-H-5440 (see 4.6.1.1).

3.3.3.1 Hydraulic reservior and lines. The internal cleanliness of the hydraulic reservoir, lines, and fittings shall be maintained throughout the installation of the hydraulic system (see 4.6.1.1).

3.3.3.2 Hydraulic fluid leakage. The loader hydraulic system specified herein shall evidence no sign of leakage through reservoir seals, gaskets, joints and manifolds while being operated at nominal operating hydraulic pressure under all conditions of operation, except at moving seals. The leakage at moving seals shall not exceed one drop per 25 cycles (see 4.6.1.3).

3.3.4 Exhaust system. There shall be no looseness or leaks in the exhaust system (see 4.6.1.4).

3.3.5 Adjustment mechanisms. All adjusting mechanisms shall be operational and capable of maintaining adjustment settings under all operating conditions (see 4.6. 1.5).

3.3.6 Latching devices. All latching devices shall be positive in fastening (see 4.6.1.6).

3.3.7 Lights. Normal headlights and tailights, blackout headlights and tailights and all instrument panel lights shall conform to applicable drawings. All electrical contacts and connections shall maintain positive contact under all loader operating conditions (see 4*6.1*6).

3.3.8 Welding and burning. Welding and burning of any type or class shall be made only when, and to the extent, specifically authorized by the acquisitioning activity (see 4.6.1.7).

3.3.9 Stowed material. Unless otherwise specified (see 6.2), all on vehicle material (OVM) shall be stowed on the vehicle in the spaces provided to assure that it will not interfere with the operation of the vehicle and/or components in any manner. For stowing purposes, the manufacturer installed OVM as furnished by the vehicle contractor, and the depot installed OVM as furnished by the Government, shall be of the latest configuration and production (see 4.6.2.1).

3.3.10 Structural interferences. All moving joints, fittings, hoses, cables, and structural members shall be free from detrimental interferences (see 4.6.1.8).

3.3.11 Missile transport. The loader shall transport 1,2, or 3 missiles to whatever location is necessary within a Hawk missile system battery commensurate with the applicable missile latching points (see 4.6.2.2).

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3.3.12 Lifting and towing. The loader shall operate as a general lifting and towing device at a battery site, except as restricted by the operational limitations of the loader as specified herein (see 4.6.3.3).

3.3.13 Superstructure alignment. With the index boom properly indexed to the pallet or launcher, the loader hoisting beam latches shall be in proper alignment with the applicable missile latching points (see 4.6.1.9).

3.4 Ike-test break-in run (all vehicles).

3.4.1 Preparation. Prior to pre-test run, the loader shall be completely assembled and all adjustments completed. The loader, including all components requiring lubrication shall be lubricated and serviced as required, using standard military products as shown on the lubrication drawings (see 4.4.3).

3.4.2 Engine warm-up. Prior to start of pre-test run, or any time during the the run when the engine has been stopped for a period of 15 minutes, the engine, when started, shall be warmed-up for at least 3 minutes and all engine indicator lights and gages shall indicate normal operating conditions (see 4.4.3).

3.4.3 Distance operated. Each loader, without missiles and with the superstructure in normal Transport position, shall be given a pre-test run on smooth, level, hard-surfaced roads, except a section of the road shall be as shown in Figure 1. The loader shall be operated as specified in table I, and over the section of road shown in Figure 1 in forward and reverse gear (see 4.4.3).

TABLE I. Speeds and distance for pre-test run.

Division of run	Engine RPM	Drive-selector position	Distance yards	Distance (meters)
a	1500	H	880	(805)
b	2200	H	880	(805)
c	2000	R	16.6	(15.2)
d	1500	D	880	(805)
e	2400	D	880	(805)
f	2000	R	16.6	(15.2)

Note: H - Hilly, D - Drive, R - Reverse

3.4.4 Reverse operation. Before operating in reverse (R), each time during the pretest run, the loader shall be brought to a stops placed in neutral and the engine allowed to idle for not less than two minutes (see 4.4.3)*

3.4.5 Condition after pre-test. After completion of the pretest break-in run, there shall be no excessive grinding or backlash in gear trains, and all suspension wheel bearings shall be correct or readjusted as required. There shall be no indication of damaged components, or maladjustments that may cause faulty vehicle operation (see 4.4.3).

a

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3.5 Performance. A complete loader, serviced with specified fuel, lubricants, grease, and fluids, and unless otherwise specified, without intended or simulated load, shall perform as specified herein. Vehicle, serviced (see 3.4.1) and equipped for existing climatic conditions, shall operate as specified without special equipment. The loader shall operate without evidence of excessive vibration, overheating of engine, power train, and assemblies. There shall be no evidence of leaking of lubricants, water, fuel, or hydraulic fluids, looseness, misalignment, misadjustment, binding of parts, or any other malfunction during operation. Unless otherwise specified, the loader shall demonstrate performance requirement on relatively smooth, level, hard-surfaced roads (see 4.6.1.10).

3.5.1 Loader mobility. Unless otherwise specified, the loader, with the loader, with superstructure in transport position and properly serviced, shall meet the following requirements (when adequate surface friction is available) under all operating and environmental conditions specified herein. The loader shall operate without evidence of excessive vibration, overheating of the engine, power train or suspension system assemblies. There shall be no evidence of leaking of lubricants, water, fuel or hydraulic fluids, loseness, misalignment, misadjustment, binding of parts; or any other defect or malfunction during operation as specified.

3.5.2 Power package. When the power package is installed in accordance with applicable drawings, it shall function throughout all gear and speed ranges without loss of lubricants or damage that may cause failure of the loader (see 4.6.1.10.1).

3.5.3 Power train. The power train and associated controls shall operate throughout all speed and steering ranges without binding of linkages, loss of lubricants , grabbing, and chattering or slippage when controls are applied to control, stop, and hold the vehicle (see 4.6.1.10.1).

3.5.4 Speed. With drive selector in drive (D) position, the loader shall start out with the transmission in low range, then shift to transmission intermediate range, then shift to transmission high range, and shall maintain a speed of not less than 10 miles per hour (mph) [16 kilometers per hour (km/h)] for extended periods of time. With drive selector in hilly (H) position, loader shall operate at sustained speeds up to 5 mph (8 km/h) without transmission shifting (see 4.6.1.10.2).

3.5.5 Creeping. With the drive selector in D position, and utilizing brakes as required, the loader shall be capable of smooth creeping at speed not to exceed 6 inches [15 centimeters (cm)] per second (see 4.6.1.10.3).

3.5.6 Acceleration. The loader shall accelerate from a standing start, with engine at idle, to 10 mph (16 km/h) in not more than 45 seconds (see 4.6.1.10.4).

3.5.7 Climbing. The loaded vehicle shall climb, in both forward and reverse gear, a relatively smooth longitudinal 40-percent grade from a standing start on the grade. The loader without missiles, shall be capable of the same performance on a relatively smooth 60-percent longitudinal grade (see 4.6.1.10.5).

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3.5.8 Braking.

3.5.8.1 Stopping. The loader, traveling at 10 mph (16 ksn/h) shall stop within 10 feet from the point of brake application (see 4.6.1.10.6).

3.5.8.2 Holding. With the loader weighted with the maximum load or a simulated load of equal size and weight, and on a 40 percent longitudinal slope, the service brakes shall hold the vehicle stationary for not less than 3 minutes, when headed up and down grade. With brakes locked for parking and all other holding devices inoperative, the loader shall be held stationary when headed up and down grade. On a relatively smooth, level, hard surface, the loader shall not move when the engine is accelerated to a minimum of 1,000 rpm, with the drive selector in H position and both brakes fully applied (see 4.6.I.10.7).

3.5.9 Turning. The loader shall turn to the right and to the left, in forward or reverse, full 360-degree turns with the carrier remaining within a circle 40 feet [12*2 meters (m)] in diameter (see 4.6.1.10.8).

3.5.9.1 Brake lever force. Brake lever force necessary to start the turn at all temperatures down to 32 degrees Fahrenheit (°F) [0 degrees Celsius (°C)], shall not be more than 40 pounds [177.9 newtons (N)], and the lever force necessary to hold the turn shall be not more than 30 pounds (133.5 N). The increase in lever force at all temperatures between 32°F (0°C) and minus 65°F (-18°C) shall not exceed 20 percent. With all brakes released and steering levers in the full forward position (for "straight-ahead" or "straight-back' loader movement) and left untouched, the loader shall maintain a relatively straight course when operated in the forward and reverse direction. When traveling at a speed between 5 and 10 mph (8 and 16 km/h), drift shall not to exceed 3 feet in 100 feet (0.91 m in 30.5 m) (see 4.6.3.4).

3.5.10 Side slope. The loader shall transport at maximum speed a complement of three missiles on a smooth 30 percent side Slope. With less than three missiles and the load unbalanced in a downhill direction, the loader shall operate at maximum speed on a relatively smooth 18 percent side slope (see 4.6.2.3).

3.5.11 Trench crossing. The loader shall cross trenches up to 25 inches (63.5 cm) wide at reduced speed (see 4.6.3.5).

3.5.12 Vertical obstacle. The loader shall cross over vertical obstacles up to 10 inches (25 cm) high at reduced speeds (see 4.6.3.6).

3.5.13 Blast hole. The loader shall negotiate a generally circular blast hole of 5 feet (1.52 m) in diameter and 18 inches (45.7 cm) in depth at reduced speed (see 4.6.3.7).

3.5.14 Rough terrain. The loader shall negotiate hard, boulder-strewn terrain, with an obstacle height of no more than 5 inches (12.7 cm), at speeds up to 5 mph (8 km/h). The loader shall negotiate undulating terrain (see figure 1) at speeds up to 10 mph (16 km/h) (see 4.6.3.8).

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3.5.15 Operation in snow. The loader when operating for extended periods of time in wet snow up to 12 inches (30 cm) in depth shall meet performance requirements specified herein (see 4.6.3.9).

3.5.16 Fording. The loader shall ford a salt or fresh water crossing up to 20 inches in depth, including sinkage and wave height, without the use of special equipment (see 4.6.3.10).

3.5.17 Towing. The loader, without missiles, shall tow a load developing a=abay drawbar pull of not less than 5,000 pounds [2,268 kilograms (kg)] and a continuous drawbar pull of 2500 pounds (1134 kg) for a distance of not less than 1,000 feet (304.8 m) at a minimum. speed of not less than 2 mph (3.22 km/h) over relatively level, smooth, and firm terrain (see 4.6.3.11).

3.5.18 Loader missile handling.

3.5.18.1 Hydraulic pressure at loader engine start-up. With the starter switch in the engine starting position and the engine turning over, the hydraulic pressure registered on the loader hydraulic oil pressure guage shall not exceed 50 psi [344.7 kiloPascals (kPa)] when the hydraulic load is removed from the engine. With the engine running and the starter switch released, (normal loader operating position), hydraulic oil pressure shall be as specified in 3.5.20 (see 4.6.1.12).

3.5.18.2 Loader electrical systeme The loader electrical system shall operate from a regulated 24 volt direct current (DC) supply and shall perform as specified herein under the range of environmental and service conditions to which the loader is exposed during operation. All electrical components shall be securely mounted-electrical connections shall be tight, and externally exposed connectors shall have watertight seals. Cabling shall show no evidence of broken insulation, broken strands, fraying or detrimental rubbing against other loader parts (see 4.6.1.12.* a

3.5.18.3 Master switch. With the master switch “OFF”, no electrical power shall be provided to operate the loader electrical system. With the master switch “ON” , power shall be supplied to the electrical system and the carrier and superstructure circuits shall operate as specified herein (see 4.6.1.12).

3.5*18.4 Ignition switch. With the ignition switch at “ON” position, engine starting and operating circuits, engine indicator lights, gages, missile latch and release indicator lights and fuel pump shall be operable, With the switch at “OFF” position, the operating circuits and fuel pump shall be inoperative (see 4.6.1.12).

3.5.18.5 Engine starting circuit. With the loader in operating condition, properly serviced, and with the drive selector in “N” position, the starting circuit shall function as follows when the starter switch is operated as follows (see 4.6.1*12):

a. With the ignition switch “OFF” and the starter switch in the engine starting position, the starter motor shall operate, but the engine shall not start.

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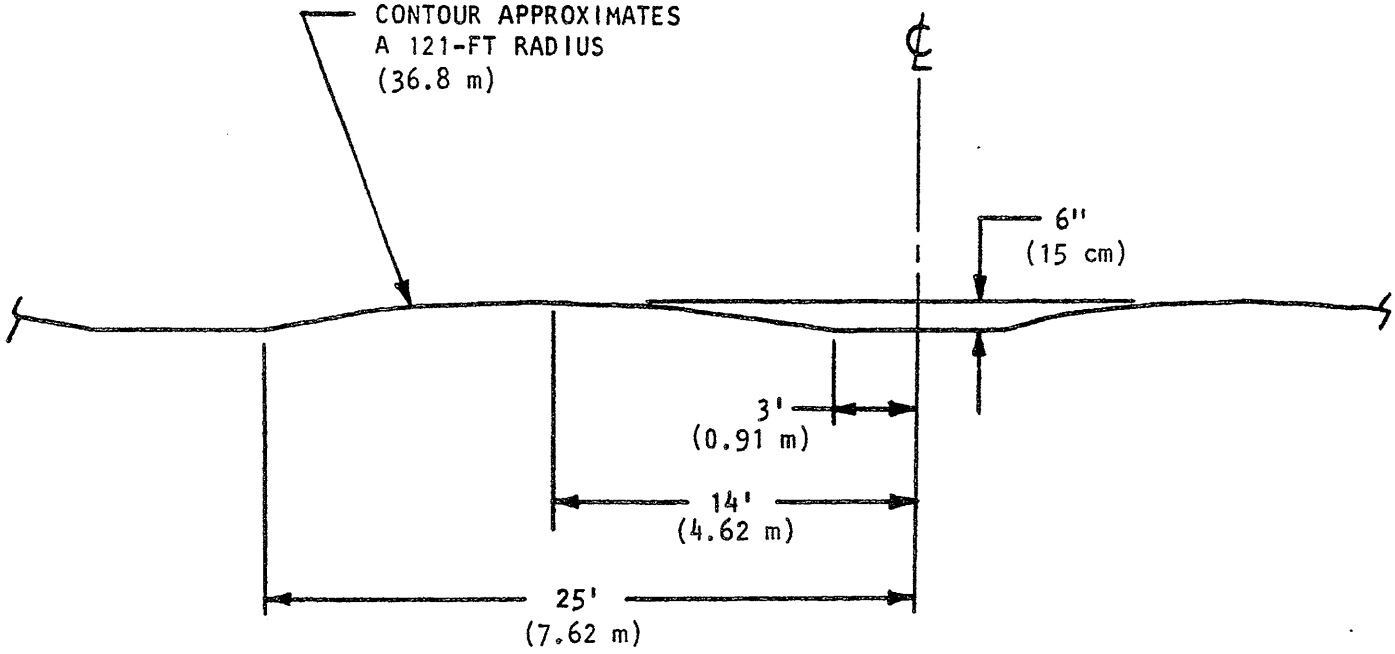


FIGURE 1. Profile of Test Track (Artificial Bump).

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be With the ignition switch at "ON" and the starter switch in the engine starting position, the starter motor shall operate, and the engine shall start.

3.5.18.6 Lighting circuits. With light switch assembly on the instrument panel, adjusted as shown in table II, lights shall operate as specified herein (see 4.6.1.12).

TABLE II. Instrument panel lights.

Switch setting and operating conditions	Light conditions
1. Main light switch at "off"	No carrier lights operate.
2. Main light switch at "service drive"	Head and tail lights on.
a. Dimmer switch operated	Headlight beam changes.
b. Brakes applied	Stoplight on.
c. Brakes released	Stop light off.
3. Main light switch at "stop light"	Head lights off, tail lights off.
a. Brakes applied	Stop light on.
b. Brakes released	Stop light off.
4. Main light switch at blackout (bo) marker	Bo marker and bo tail lights on.
a. Brakes applied	Bo stop light on.
b. Brakes released	Bo stop light off.
5. Main light switch at (bo) drive	Bo drive light on bo marker and bo tail lights on.
a. Brakes applied	Bo stop light on.
b. Brakes released	Bo stop light off.
6. Main light switch at any setting except "Off" and panel light switch at settings as follows:	Lights operate as specified in 2, 3, 4, or 5 as applicable.
a. "Off" position	Instrument panel lights off.
b. Park	Instrument panel lights on.
c. Dim	Instrument panel lights dim.
d. Panel bright (brt)	Instrument panel lights bright.

3.5.18.7 Safety switch. With the safety switch "ON", and the light switch assembly in normal position, the main light switch (see table 11) shall be operable only in the "OFF" or "be" marker position. The main light switch shall be turnable to "service drive", or "stop light" positions as specified in table 11 only when the safety switch is simultaneously held in unlock position (see 4.6.1.12).

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3.5.18.8 Engine indicator circuits. Engine indicator gages and lights shall indicate as shown in table III (see 4.6.1.12).

TABLE III. Indicator gages and lights.

Indicator gage or light	Indication
a. With ignition switch "OFF"	
1. Battery generator (BATT-GEN) indicator	Actual battery voltage.
2. Other gages and indicator lights	Lights off gages do not register.
b. With ignition switch "ON" - engine not running	
1. (Batt-gen) indicator	Actual battery voltage.
2. Fuel level gage	Correct quantity of fuel.
3. Engine and transmission (eng and trans) low oil pressure warning lights	On.
4. Transmission, differential, hydraulic, (trans, diff, hyd) and water high temperature (temp) warning lights	Off (unless oil, hydraulic or water in high temperature range, or unless indicator lights test switch is depressed.
5. Indicator lights test switch actuated	All indicator warning lights on.
c. Engine running at fast idle	
1. (BATT-GEN) indicator	Actual generator (voltage regulator) voltage.
2. Fuel level gage	Correct quantity of fuel.
3. All indicator warning lights	Off, except as specified in (b) 4 above.
4. Indicator lights test switch actuated.	All indicator warning lights on.

3.5.18.9 Auxiliary power outlet. Auxiliary 24 volt DC power fused at amperes (amps) shall be provided at the utility outlet (see 4.6.1.12).

3.5.19 Controls.

3.5.19.l Engine, transmission and electrical equipment controle The engines transmission, and electrical controls shall be capable of positive operation at all times and under all operating conditions (see 4.6.l.13)e

3.5.19.2 Suspension system lockout controls. With the suspension lockout control in lockout position, the right and left front and cen~er wheels of the suspension system shall be in lockout and the loader shall load and unload a pallet or operate as a crane. With the controls returned to the unlocked position the suspension system shall return to normal operation (see 4.6.l.13).

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NOTE : Suspension lockout control shall be engaged or disengaged with superstructure in transport position.

3.5.19.3 Brake controls. With the brake control levers fully released or locked, the tops of the brake lever handles shall be in alignment with each other within one-half inch. The force required to lock the brakes shall not vary between the two levers by more than five pounds. With each brake lever in fully locked position, there shall be at least two notches of travel remaining (see 4.6.1.13).

3.5.19.4 Superstructure controls. With the superstructure manual operating controls in "NEUTRAL" position, the superstructure shall remain approximately as positioned. Operation of the controls shall provide smooth positive actuation of the superstructure throughout all ranges of azimuth, elevation, roll, and extension, under all operational conditions (see 4.6.1.13).

3.5.19.5 Engine starting lockout control. With the drive selector lever in any position other than neutral (N) the engine starter electrical circuit shall be inoperative (see 4.6.1.13).

3.5.20 Hydraulic system pressure.

3.5.20.1 Hydraulic pump. The operating pressure of the engine driven hydraulic pump shall be 3,000 plus 200, minus 100 pounds per square inch gage (psig) [20.7 + 1.38, -0.7 MegaPascals (MPa)] when using oil conforming to MIL-H-5606. The maximum system pressure shall be limited to 3,400 psig (23.4 MPa), by a system relief valve (see 4.6.1.14).

3*5.20.2 Proof pressure. The hydraulic system shall withstand a proof pressures as specified in MIL-H-5440 without damage (see 4.6.1.14).

3.5.21 Superstructure operation (normal loader operation). The loader superstructure, empty or loaded with 1, 2, or 3 missiles, shall operate smoothly and satisfactorily under operating conditons as specified herein. With the superstructure adjusted for normal loader operation as speci~ied in table IV and with the engine running, the superstructure shall respond as follows when the superstructure control levers are moved from their respective neutral positions as specified herein (see 4.6.1.15).

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TABLE IV. Normal loader operation.

Lever and lever motion	Superstructure movement	
	Motion	Range of travel
a. ELEVATION - ROLL		
1. Forward and back	Index boom respectively depresses and elevates	700 mils [17.78 millimeter (mm)] (minimum) of elevation and depression.
2. Right or left (Control centers when released)	Index boom rolls respectively right or left	267 mils (6.78 mm) (minimum) in either direction from from center level position, except as limited by 3.5.21.3
b. AZIMUTH		
1. Clockwise or counterclockwise (Control centers when released)	Main support rotates in azimuth respectively clockwise or counterclockwise	267 mils (6.78 mm) (minimum) in either direction from center position, except as limited by 3.5.21.4.
c. EXTENSION		
1. Forward or back	Combined link assembly and transfer arms respectively extend and retract	Not applicable

3.5.21.1 Missile release circuits. The hoisting beam latch release solenoid shall not operate when the missile latch release switch located on the instrument box is in the normal(open) position (see 4.6.1.15.1).

3.5.21.2 Superstructure operating speeds. With the loader on a relatively level, smooth, firm surface, control levers at maximum settings, and engine speed as specified in table V, the superstructure, loaded with three missiles, shall operate as specified in the table (see 4.6.1.16).

3.5.21.3 Superstructure settling. With three missiles on the loader, elevation and extension cyclinder extended to any position within their respective ranges, elevation and extension valves in neutral, and with the engine off, the superstructure shall not settle more than one inch (measured vertically at missiles) in one minute. (Caution: If the transfer arms are raised more than 15 degrees, the loader must be engaged with the pallet or launcher to prevent overturn.) (see 4.6.1.17)

3.5.21.4 Azimuth and roll control limitations. Azimuth and roll controls shall be operative when the missile is more than 10 inches (25.4 cm) above the transport position, and shall be inoperative when the missile is less than 2 inches (5.1 cm) above the transport position (see 4.6.1.18).

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TABLE V. Superstructure operating speeds.

Operation	Engine RPM	Traverse Time (entire range) seconds
a. ELEVATION CYLINDER		
1. Elevate	2,000 - 2,500	10.0 to 21.0
2. Depress	2,000 - 2,500	10.0 to 21.0
b. ROLL CYLINDER		
1. Full right to full left	2,000 - 2,500	15.0 to 21.0
2. Full left to full right	2,000 - 2,500	15.0 to 21.0
c. AZIMUTH CYLINDER		
1. Rotate from full right to full left	1,500 - 2,000	4.0 to 12.0
2. Rotate from full left to full right		
d. EXTENSION CYLINDER		
1. Extend (loader indexed to pallet on ground	3,000 - 3,200	*15.0 to 30.0
2. Retract (loader indexed to pallet on ground)	3,000 - 3,200	*15.0 to 35.0
*Entire range includes reduced pressure travel of missiles on pallet.		

3.5.21.5 Reduced Pressure control. Extension and retraction of the superstructure during the final phase of extension and the initial phase of retraction, shall be accomplished under reduced hydraulic pressure not to exceed 630 psi by energizing the solenoid valve in the hydraulic system extension control loop. The solenoid valve shall energize under conditions (a) and (b) below, and de-energize under condition (c) and remain energized under condition (d).

(a) Index and transfer arm limit switches are actuated (plungers extended).

(b) During extension.

1. Without missiles, the solenoid valve shall energize at least 0.5 inch (1.27 cm) before the loader hoisting beam forward snap-in latch seats in the upper aft missile pocket, and not before the transfer arm separates from the intermediate link on the limit switch side.

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2. With missiles, the solenoid valve shall energize at least 0.5 inch (1.27 cm) before the applicable lower aft missile pocket seats against the aft pallet hook, and not before the transfer arm separates from the intermediate link on the limit switch side.

(c) During retraction.

1. Without missiles, the solenoid valve shall de-energize after the loader hoisting beam forward snap-in latch has moved at least 0.5 inch (1.27 cm) from the upper aft missile pocket and prior to closure of the gap between the transfer arm and the intermediate link on the limit switch side.
2. With missiles, the solenoid valve shall de-energize after the lower aft missile pocket has moved at least 0.5 inch (1.27 cm) from the aft pallet hook and prior to closure of the gap between the transfer arm and the intermediate link on the limit switch side.

(d) During extension and retraction.

1. With any combination of 1, 2, or 3 missiles and with the missile(s) latched to the pallet and to the loader simultaneously, the solenoid valve shall remain energized and there shall be no permanent deformation or damage to the loader when the extension control lever is operated in the extended direction (see 4.6.1.19).

NOTE : Solenoid valves in the roll and elevation systems shall cause the roll and elevation cylinders to become inoperative during the reduced pressure condition.

3.5.21.6 Manual latching of missiles. Positive latching and locking of 1, 2, and 3 missiles to the superstructure hoisting beams shall be accomplished by manually operating applicable latching handles to the latched position. In the latched position, missile latches shall fully extend and remain locked until the rear hoisting beam latches are retracted by manual or electrical release (see 4.6.1.20).

3.5.21.7 Manual release of missiles. Latched missiles shall release from the superstructure hoisting beams when the applicable latch release solenoid plunger is manually depressed (retracting aft latch), and when missiles are at rest on the pallet saddles so the weight of the missiles shall be completely off the hoisting beams. In the released condition, aft latches shall be fully retracted (see 4.6.1.21).

3.5.21.8 Forward missile supports. Outboard forward missile supports shall erect from the stowed position (at rest on loader fenders) to the normal loader position (from support of missiles during loader transport) by rotating the support lever handles from the full forward positions to full back and vice versa (see 4.6.1.22).

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3.5.21.9 Rear center missile support. The" rear center missile support and safety bar shall fold down to the stowing position for loader shipment and for crane operation (see 4.6.1.13).

3.5.22 Superstructure electrical system.

3.5.22.1 Main electrical cable. With the main electrical cable assembly connected between the superstructure and distribution box for normal loader operation, superstructure electrical circuits shall operate as specified herein. With the superstructure cable disconnected aid attached-to dummy a connector, the superstructure electrical circuits shall not function (see 4.6.1.24).

3.5.22.2 Superstructure circuits (normal loader operation). With the superstructure adjusted for normal loader operation with missiles, and with the master switch "ON", the superstructure circuits shall function in accordance with paragraphs 3.5.22.3 through 3.5.22.7 (see 4.6.1.24].

3.5.22.3 Missile latch and release circuits. With the ignition switch "OFF", the "missile-latched" and "missile released" indicator lights shall not illuminate. With the ignition switch "ON" and with the loader engine and superstructure operating or at rest, "snissile-re-leased" and "missile-latched" lights, the indicator-lights-test switch and dim bright control shall function in accordance with paragraph 3.5.22.3.1 through 3.5.22.7 (see 4.6.1,24).

3.5.22.3.1 Missile latch and release lights. Unless otherwise specified, the right, center or left "missile-released" (red) shall illuminate when missile latches in the corresponding (right, center or left) hoisting beam assembly are in the released condition. Right, center or left "missile-latched" light (green) shall illuminate when missile latches in the corresponding hoisting beam assembly are in latched condition (see 4.6.1.24).

3.5.22.4 Indicator lights test switch. All "missile released" and "missile latched" lights, whether individually "ON" or "OFF", shall illuminate when the "indicator-lights-test switch" is actuated (switch button held in full depressed position). Indicator lights shall return to their initial condition when the switch button is released (switch returned to normal "OFF'* operating position) (see 4.6.1.24).

3.5.22.5 Dim bright control. Rotation of dim-bright control from "dim" (maximum counterclockwise position) to "brt" (maximum clockwise position) shall brighten all illuminated "missile-released" and "missile latched" lights and vice versa. With the control at "dim", missile-released lights shall be readily visible in the illuminated condition (red), during complete darkness. With the control at "brt", illumination of "missile-released" lights shall be readily visible in bright sunlight (see 4.6.1.24).

3.5.22.6 Electrical release of missiles. With 1, 2 or 3 missiles latched to the loader, operation of missile "latch release" to momentary "ON" . (up) position shall electrically release the one or snore selected missiles from the loader when the following conditions are met (see 4.6.1.25):

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(a) Conditions for energizing latch release solenoids applicable to missiles selected for release shall be as specified herein.
(b) Weight of missiles shall be off the loader hoisting beam assembly housings (full weight of missiles carried by pallet saddles or other surface receiving released missiles)*

3.5.22.6.1 Latch release solenoids. With the loader hoisting beam assemblies in the "missile-latched" condition as specified herein, energizing the "latch release" solenoids shall retract the latch locks so that the spring-loaded latching assemblies will return to the "missile released" condition when the weight of the missiles is not suspended from the Iatchese Solenoids shall energize only when the following conditions are met (see 4.6.1.25.1):

Master switch "ON"

(a) Index and transfer arm limit switchs are actuated (plunger extended).
(b) Applicable circuit breakers are on (toggle "up" position).
(c) Missile "latch release" switch is actuated to "ON".

3.5.22.7 Index boom and transfer arm limit switches. Latch and release solenoids shall not energize unless the series connected index boom-intermediate link limit switch and the transfer arm-intermediate link limit switch are actuated (plungers extended). The index boom-intermediate link limit switch shall actuate only when the superstructure is extended until the link assembly is raised from the index boom trunnion beyond a point where the limit-switch plunger is extended. Transfer arm-intermediate link Limit switch shall actuate only when the transfer arm is separated from the intermediate link assembly beyond a point where the limit-switch plunger is extended (see 4.6.1.26).

3.5.23 Loading and unloading missiles. The loader shall load and unload missiles under the following conditions (see 4.6.3.12)~

(a) With the launcher positioned on a slope up to 10 degrees and with the loader on the same slope, the loader shall be capable of indexing to the launcher and loading or unloading the the launcher with 1, 2, or 3 missiles. When loading or unloading less than 3 missiles, the missiles may be guided by hand if necessary.

(b) With the pallet mounted on an XM-390 type trailer, positioned on slopes up to 10 degrees and with the loader on the same slope, the loader shall be capable of indexing to the pallet and loading and unloading the pallet with 1, 2, or 3 missiles~ When loading or unloading less than 3 missiles, the missiles may be guided by hand if necessary~

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NOTE : The trailer mounted pallet must be approximately leveled when the trailer pintle is down slope or when located on a side slope. When the trailer pintle is up slope the trailer bed must be approximately parallel to the ground.

3.5.24 Loading and unloading pallet. With the suspension locked, the loader, either empty or with missiles aboard, shall be capable of engaging an empty pallet and transferring the pallet from an XM-390 type trailer to the ground, and from the ground to an empty XM-390 type trailer (see 4.6.1.28).

3.5.24.1 Proof load. the loader shall be capable of withstanding loads imposed by 110 percent of the normal improved missile weight [1526 to 1580 lbs (6,92.2 to 716.7 kg)] during dynamic transfer operations.

3.5.25 Crane.

3*5.25.1 Crane operation. With the suspension locked, the loader, when titted with crane accessories and adjusted for crane operation, shall function as a crane on slopes up to 10 degrees commensurate with the loader crane operating requirements as specified herein. All moving joints, fittings, hoses, electrical cables, stays, and structural members shall be free from binding or detrimental interference during operation (see 4.6.3.13).

3.5.25.2 Crane controls. With the loader engine operating, the speed and direction of cable travel and swing and elevation of the crane boom shall be controlled by operation of the superstructure hydraulic controls specified herein (see 4.6.3.13.1).

3.5.25.3 Roll lockout. With the roll lockout lever placed in “crane” position, roll control shall be inoperative (see 4.6.3.13.2).

3.5.25.4 Hoisting cable. Operation of extension control lever forward and backward shall respectively raise and lower the maximum loads as specified herein (see 4.6.3.13.3).

3.5,25.5 Boom swing. With maximum loads as specified herein applied to boom, operation of the ”azimuth” control lever clockwise and counterclockwise shall respectively swing the crane boom (ramp beam) clockwise and counterclockwise (see 4.6.3.13.4).

3.5.25.6 Boom elevation. With maximum loads as specified herein applied to the boom, operation of the elevation control lever forward and back shall respectively elevate and depress the crane boom (see 4.6.3.13.5).

3.5.25.7 Hoisting hook position. With the crane rigged without cable stay and load, the hoisting hook shall be positioned (relative to the ground) as specified as follows (see 4.6.3.13.6):

(a) With the crane boom (ramp beam) fully depressed (elevation cylinder fully extended) and the transfer arm sockets seated on the index boom trunnion (hoisting cable Fully extended), the lowest point of the hoisting hook shall not be higher than 12 inches above the ground.

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(b) With the crane boom fully elevated (index boom resting on the rear missile support) and the extension cylinder fully extended (hoisting cable fully retracted) the lowest point of the hoisting hook shall be not lower than 150 inches (381 cm) above the ground.

3.5.25.8 Rated capacity. The loaders in a non-mobile condition on slopes up to 10 degrees, shall maintain stability when lifting a minimum load of 2,500 pounds (1,134 kg). The loader in a mobile condition on slopes up to 10 degrees (operation at reduced speed over unimproved terrain) shall maintain stability when lifting a load of 1,600 pounds (725.7 kg). When operated at either condition, there shall be no permanent deformation or damage to the loader (see 4.6.3.13.7).

3.5.26 Radio interference suppression. The complete loader shall be suppressed to prevent interference to high-sensitivity radio communication equipment in the frequency range of 0.15 megacycles (me) to 500 mc, in accordance with the applicable provisions of MIL-E-55301 for tactical vehicles (see 4.6.3.14).

3.6 Weight. The weight of the completely assembled loader, serviced with hydraulic fluid conforming to MIL-H-5606 but excluding missiles, fuel, storage items, crane installation and an operator, shall not exceed 5,500 pounds (2,495 kg) (see 4.5.1,2).

3.7 Painting, marking and data plates.

3.7.1 Painting. Unless otherwise specified, the exterior and interior of loaders components, assemblies subassemblies, and parts that require painting shall be prepared and painted in accordance with MIL-STD-193 and applicable drawings.

3.7.2 Marking. Marking shall be in accordance with MIL-STD-642.

3.7.3 Name, shipping and indentificati.on data plates. The information to be included shall be in accordance with applicable drawings or as specified by the procuring activity. The plates shall meet the performance requirements of MIL-P-514.

3.8 Fire extinguisher. When mounted in accordance with the drawings, the extinguisher shall be readily accessible and shall operate without malfunction.

3.9 Workmanship. The workmanship employed in the manufacture of the loader shall be of such quality and consistency as to minimize degradation of Ehe expected performance and design reliability. The loader shall be free from such defects as breaks, bends, cracks, deep cuts , rust, chipped or spotty paint, damaged fastenings, looseness of attaching parts, misalignment and binding or detrimental interference of parts. All bolted joints shall have reasonably flat mating surfaces that are free from burrs, weld spatters and other significant surface irregularities.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractorfs quality control system. Unless otherwise specified by the procuring activity, the contractor shall provide and maintain an effective Inspection and Quality Control System acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced therein.

4.1.2 Parts and components. Parts, components and assembles shall be inspected in accordance with the applicable drawings and Quality Assurance Provisions (see 3.3).

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Inspection equipment. Inspection equipment drawings referenced ‘by number herein are referenced to Drawing 11675300. Inspection equipment drawings so referenced, will serve as standards for determining the adequacy of contractor s inspection equipment (see 6.5).

4.1.5 Qualified products. The contractor s inspection records shall be checked to determine contractor conformance to 3.2.1.

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4.2 Classification of inspections. Vehicle inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).
c. Acceptance test (see 4.6 - 4.601.28).
d. Control test (see 4.6.2 - 4.6.2.5).
e. Comparison test (see 4.6.3 - 4.6.3.16).

4.3 Inspection conditions. Unless otherwise specified, all vehicle inspections shall be performed in accordance with test conditions specified in paragraphs 4.4.1 through 4.4.2.3 of this specification-

4.4 First article inspection. First article inspections are categorized as preproduction inspection or initial production inspection. "Approval of the first article sample by the Government shall not relieve the contractor of the obliga~ion to supply loader-transporters (XM501E3) that are fully representative of those inspected as a first article sample Any changes or deviation of the production units from the first article sample shall be subject of the approval of the contracting officer.

4.4.1 3 Preproduction inspection. When specified (see 6.2), one preproduction vehicle shall be inspected for conformance to all requirements contained in this specifications The inspection shall be performed at a location approved by the Government*

4.4.1.1 Preproduction inspection failure. Failure of the preproduction vehicle to comply with any of Che requirements specified herein shall 'be cause for refusal by the Government to conduct a retest until corrective taken by the contractor has been approved by the Government.

4.4.2 Initial production inspection. Unless otherwise specified (see 6*2), initial production vehicles shall be subjected to a special process inspections conducted by the Government representatives, during fabrication of the first production vehicle (see 3.2) to determine conformance of materials and workmanship with drawing and specification requirements After completion of the first vehicle inspections a second vehicle selected from the first months production, or one of the first 10 vehicles produced shall be subjected by the Government to examination and tests specified in 4.2 (c) and (d). Subsequent to examination and tests, the selected vehicle shall be subjected by the Government to the 250 hours test specified in table VI at a site approved by the Government, which will require a maximum of 60 days. Delays caused by vehicle breakdown due to poor quality of workmanship or material, or failure of the contractor to comply with specifications or drawing requirements shall not be the basis for adjustment of the contract performance dates or delivery schedule or of the contract price.

4.4.2.1 Initial production test failures. Failure of the vehicle to comply with any of the requirements specified or any deficiency of a workmanship or materials nature found on the vehicle during or as a result of the 250 hour test, shall be cause for rejection of the vehicle. The Government shall refuse to continue acceptance of production vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. In addition, correction of the deficiency

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must be accomplished by the contractor at no cost to the Government. Any deficiency found as a result of the 250 hour test shall be considered prima facie evidence that all vehicles already accepted prior to completion of the 250 hour test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies shall be corrected by the contractor at no cost to the Government.

TABLE VI. 250 Hour test.

A. Cycle breakdown for loader operation	
1. Missile transfer	70 hours
2. Missile transport	70 hours
3. Miscellaneous	60 hours
B. Cycle breakdown for mobile crane	
1. Load handling	12 hours
2. Maneuvering	12 hours
3* Miscellaneous	12 hours
C. Cycle breakdown for use as utility vehicle	
1. Maneuvering	6 hours
2. Hauling extremely heavy load	3 hours
3. Miscellaneous	5 hours

4.4.3 Pre-test break-in run. As specified in 3.4.1, 3.4.2, and 3.4.3 each loader shall be lubricated and serviced prior to the pre-test run. The run shall be on smooth, level, hard-surface roads, and for the distances specified in table 1. The loader shall be operated in reverse after each division as specified in 3.4.4. The undulating portion of the test shall be accomplished twice in each direction during division "a" and during division "b"

4.4.3.1 Condition after run. After the pre-test run the loader shall be run at idle speed for a period of not less than 5 minutes, and inspected to determine conformance to 3.4.5.

4.4.4 Quality conformance records. Review and evaluation of records for processing procedure, inspection systems and plan of quality control, will be conducted on materials welding, surface hardening, heat treating, protective finishes for all first article initial production vehicles.

4.4.5 Hull inspection. The hull shall be inspected prior to covering the weldments with paint or other pertinent components or assemblies.

4.4.6 First vehicle inspection. The first vehicle when complete, shall be inspected by the Government subsquent to contractors inspection at the contractor's plant to determine conformance to contract and specification requirements.

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4.5 Quality conformance inspection.

4.5.1 Lot size. An inspection lot shall consist of all vehicles of one types from one day's production submitted at one time for acceptance inspection.

4.5.1.1 Sampling for inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected, from each inspection lot in accordance with inspection level 11 of Standard MIL-STD-105. Before sampling may be accomplished by the contractor. A process average shall be established (see 6.6) to allow normal sampling in accordance with MIL-STD-105.

4.5.1.2 Weight measurement. To determine conformance to 3.6, sampling for weight measurement of the loader shall be conducted on one loader for each lot produced.

4.5.2 Examinations* Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance *to* the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in table VII. The following constitutes a part of the classification of defects (see 6.7):

(a) by drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
(b) Any droplet that occurs a~ a static fit, metal to metal or gasket combination shall constitute a major defect.

TABLE VII. Classification of defects.

Characteristics	Major	AQL 25 Defects/100 units	Method of Inspection
Engine	101	Malfunction, unusual noise, leaks, (see 6.7 b&c) mounting improper, damage.	Visual and Functional
Power train	102	Malfunction, unusual noise, leaks, (see 6.7 b&c) mounting improper, damage.	Visual and Functional
Fuel system components	103	Malfunction, damage, leaks. (see 6.7a).	Visual and Functional
Fuel, and oil lines, valve and hose	104	Defective, clearance improper, malfunction leaks (see 6.7a), fuel cut off valve damage.	Visual and Functional
Exhaust system components	105	Damage, leaks.	Visual and Functional

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TABLE VII. Classification of defects. - Continued

Characteristics	Major	AQL 25 Defects/100 units	Method of Inspection
Cooling system components	106	Malfunction, damage, leaks (see 6.7d), unusual noise.	Visual and Functional
Electrical system components	107	Improper installation, loose connections and components, safety wiring missing. Wire has excessive bends, pinches, wiring damage, defective, cabling frayed, chafed, broken, burned, insulation damaged.	Visual and Functional
Suspension system components	108	Damage, improper clearance or alignment, welding defects, leaks, improper assembly, installation, bolt or nut torque and track tension.	Visual and Functional
Mating flanges	109	Surfaces not flat for mating when bolted or screwed together.	Visual
Hydraulic system	110	Hydraulic lines dented, twisted, buckled, excessive bend, chafed or excessively worn. Leakage (see 6.7a), hydraulic oil contaminated.	Visual and Functional
Brake lever	111	Top of brake levers out of alignment when released or locked. Brake lever travel incorrect when fully locked.	Visual and Functional
Quick release pins	112	Quick release pins do not fit holes, bent or damaged.	Visual and Functional
Center rear missile support	113	Will not adjust to stowed position and vice versa.	Visual and Functional
Outboard forward missile supports	114	Will not adjust from stowed position to the missile transport position and vice versa.	Visual and Functional
Angle of forward missile supports	115	Angle of missile transport supports incorrect.	Visual
Inboard edges of index tips	116	Not parallel.	Visual

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TABLE VII. Classification of defects. - Continued

Characteristics	Major	AQL 25 Defects/100 units	Method of Inspection
Centerline of hoisting beam housings	117	Not parallel and centerline of each housing not coincident with with centerline of its latches, distance between adjacent center-lines of housings.	Visual
Torque on latching handles	118	Inproper torque.	Visual and Functional
Transfer arm limit switch plunger	119	Plunger does not extend beyond striker face.	Visual
Index boom limit switch plunger	120	Improper travel of plunger.	Visual and Functional
Holes in Extension cylinder support bracket	121	Holes improper size or location.	Visual
Holes in index forks	122	Holes improper size or location.	Visual
Parts missings, improperly assembled and in-stalled, hose, damaged or defective	123	Lockwashers, locknuts, locking wires, cotter pins, screws, bolts, nuts, washers, rivets, captivating chains, wiring clips, grommets, ground straps, missing grease fittings, cracks, breaks, tears, deep cuts, dents, bends, or structural deformation.	Visual
Finish	124	Not in accordance with Military Standard.	Visual
Nameplates	125	Missing, not properly fastened to loader, incorrect or illegible.	Visual
Identification, operational, warning, and caution markings	126	Missing, incorrect, incomplete, illegible or not correctly located.	Visual
Speed	127	Nonconformance.	Functional
Creeping	128	Nonconformance.	Functional

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TABLE VII. Classification of defects. - Continued

Characteristics	Major	AQL 25 Defects/100 units	Method of Inspection
Acceleration	129	Nonconformance.	Functional
Reversing	130	Noncomformance.	Functional
Turning and braking	131	Nonconformance.	Functional
Slope operation	132	Nonconformance.	Functional
Instrumentation, switches, warning indicating, and safety devices	133	Malfunction, damage.	Visual and Functional
Missile circuit release test	134	Malfunction.	Functional
Superstructure circuit test	135	Malfunction.	Functional
Suspension system lockout-test	136	Malfunction.	Functional
Missile latches	137	Improper latching	Visual and alignment fixture
Superstructure alignment	138	Misalignment of any element the superstructure.	Alignment Fixture

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TABLE VII. Classification of defects. - Continued

Characteristics	Minor	AQL 150 Defects/100 units	Method of Inspection
Parts	201	Improperly assembled, loose, damaged or defective, also screws, bolts, nuts, washers, rivets, wiring clips, and grommets missing, incorrect, damaged, defective, or improperly installed (not affecting loader functional performance).	Visual
Defects	202	Cracks, breaks, tears, deep cuts, dents, bends, bulges, or structural deformation (not affecting loader functional performance).	Visual
	Minor	AQL 125 Defects/100 Units	
Loader not clean	203	Dirt, mud, oil, grease, metallic chips, or other loose, or attached foreign particles or substances present.	Visual
Welding	204	Warpage or of poor appearance.	Visual
Lubrication	205	Lubrication fittings not as required.	Visual
Identification, operational, warning and caution product markings	206	Damaged or not applied as specified (still useable).	Visual
Hoisting beam	207	Not striped as specified or striping damaged.	Visual
Azimuth index marks and roll index mark	208	Marks missing, illegible or incorrectly located.	Visual

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TABLE VII. Classification of defects. - Continued

Characteristics	Minor	AQL 125 Defects/100 units	Method of Inspection
Loader drain, covers	209	Damaged or defective.	Visual
Pintle assembly	210	Damaged, defective, improperly installed, improper operation.	Visual
Stowage box	211	Stowage box not as specified, damaged, latches inoperable, or wrong location.	Visual

4.5.2.1 Unclassified defectis. All defects having no ef fect on function! safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a thirty-day period will be added to the minor classification with no increase in AQL's

4.5.2.2 Recurring major deficiencies. A major deficiency (see 6.8) is recurring when the same defect occurs more than once in the same s-ample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found.

4.5.2.3 Recurring minor deficiencies. A minor deficiency (see 6.8) is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. Correction of all deficiencies shall be accompli-shed prior to acceptance.

4.5.3 Control tests. Control tests specified under 4.6.2 shall be conducted on the first loader off the production line or one of the first 10 loaders produced. 'Thereafter, one loader shall be selected each month and subjected to the specified tests. The control test samples shall 'be vehicles which have been subjected to the acceptance tests specified in 4.6.

4.5.3.1 Test failure. If a vehicle fails to pass any control test specified herein, the Government inspector shall stop acceptance of subsequent lots until objective evidence has been provided by the contractor that corrective action has been taken.

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4.5.4 Inspection comparison tests. The Government may select loaders any time during the contract production period and subject loaders to tests specified in 4.4.2.1, table VI to reveal deficiencies of a workmanship or materials nature that may reduce the effective operation of these items in the the field and to compare existing quality with previous standards. These tests and those specified under 4.6.3 shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Selection of loaders shall be on a spot check basis. Comparison test loaders shall be fully loaded with all OVM stowed in their applicable spaces to conform to 3.3.9. Vehicles selected shall not include any loader previously tested for conformance to 4.4.2.

4.5.4.1 Test failure. Failure of any loader tested as specified in 4.5.4 to comply with any of the requirements specified in the contract, or any major deficiency of a workmanship or materials nature occuring, or as a result of the test, may be cause for refusal to continue acceptance of loaders by the Government until evidence has been provided by the contractor that corrective action has been taken to eliminate the condition.

4.6 Acceptance tests. To determine conformance EO 3.5, after pre-test run (see 3.4) each loader selected in accordance with 4.5.1.1, shall be operated for a distance of 1 mile or more and shall be inspected as specified in 4.4.2. Acceptance tests shall include all acceptance tests and examinations specified in paragraph 4.6.1.l through 4.6.1.29. Loaders shall be completely assembled and serviced, with fuel, lubricants and fluid as specified in the production lubrication drawing. When specified (see 6.2), the loader shall carry a simulated missile payload. Operation shall be on smooth, level, hard-surfaced roads except the road shall include a section as shown in Figure 1, to test the functioning of suspension system and shock absorbers. Prior to actual tests the engine shall be operated between 650 and 750 rpm to warm the engine and lubricants to the operating temperature. The loader shall be stopped with the engine idling for not less than 30 seconds and then operated in reverse for not less than 50 feet (5.2 m). To facilitate examination, observation, and function of certain areas, this test may be conducted without the floor plates, personnel seat cushions, OVM and affixed decals. After tests, the loader shall be examined for evidence of fuel or lubricants leakage, or other deficiencies.

4.6.1 Test failure, If the vehicle fails to pass any acceptance test specified herein (see 4.6), the Government inspector shall stop acceptance of subsequent lots until evidence has been provided by the contractor that corrective action has been taken.

4.6.1.1 Fuel system, and hydraulic system check. To determine conformance to 3.3.2.1, 3.3.3, and 3.3.3.1, prior to initial fueling and filling, the fuel tank and hydraulic reservoir shall be checked for cleanliness.

4.6.1.2 Fuel system leakage check. To determine conformance to 3.3.2.1, prior to engine installation, the fuel system shall be checked for leakage.

4.6.1.3 Hydraulic fluid Leakage check. To determine conformance to 3.3.3.2, the hydraulic system shall be operated at operating pressure and checked for leakage.

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4.6.1.4 Exhaust system check. To determine conformance to 3.3.4, during loader operation the exhaust system shall be checked for functional requirements.

4.6.1.5 Adjustment mechanism check. To determine conformance to 3.3.5, all adjustment mechanism settings shall be checked for proper adjustment and readjustment made when necessary.

4.6.1.6 Fasteners, and light checks. To determine conformance to 3.3.6 and 3*3.7, all fasteners, locks, and lights shall be operated during and after break-in run and checked for functional requirements.

4.6.1.7 Welding and burning check. To determine conformance to 3.3.8, all welding and burning operations shall be checked for removal of slag and if approved procedures have been followed.

4.6.1.8 Structural interferences check. To determine conformance to 3.3.10, during loader operation all specified items shall be checked for detrimental interference.

4.6.1.9 Superstructure alignment check. To determine conformance to 3.3.13, the empty loader shall be indexed to the 10879601 alignment checkout fixture and superstructure shall be extended until the hoisting beams are in their latching position. Operate the hoisting beam latch levers until the latches are properly engaged in the fixture pockets. The latches shall engage pockets readily without requirement for application of external force to the hoisting beams. Operate manual latch release. Latches shall release readily without evidence of binding.

4.6.1.10 Loader performance mobility tests. To determine conformance to 3.5 and 3.5.1$_{s}$ the loader shall be operated in accordance with the tests specified and shall meet the performance requirements contained herein.

4.6.1.10.1 Power package and power train test. To determine conformance to 3.5.2 and 3.5.3, the loader shall be operated throughout all ranges of operation and loader observed for functional requirements.

4.6.1.10.2 Speed tests. To determine conformance to 3,5.4, the loader shall be operated as specified and observed for speeds attained.

4.6.1.10.3 Creeping test. To determine conformance to 3.5.5, the loader shall be operated at a speed of not more than 6 inches per second during which the vehicle shall be checked for smoothness of movement.

4.6.1.10.4 Acceleration test. To determine conformance to 3.5.6, the loader shall be accelerated, from a standing start with the engine idle, at the most rapid speed possible and the rate of speed measured.

4.6.1.10.5 Climbing test. To determine conformance to 3.5.7, the loader shall be operated as specified, on indicated grade and checked for functional requirements.

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4.6.1.10.6 Stopping test. To determine conformance to 3.5.8~1, the loader shall be checked for stopping requirements

4.6.1.10.7 Holding test. To determine conformance to 3.5.8.2, the loader with simulated load shall be stopped on a 40 percent longitudinal slope and the holding capability of the service and parking brakes measured when heading up and down grade. The holding capability of the brakes will be measured with the loader operating on a smooth, level, hard surfaces the engine accelerating at no less than 1000 rpm, the drive selector in the H position and both brakes fully applied.

4.6.1.10.8 Turning test. To determine conformance to 3.5.9, the loader shall be operated, in a forward or reverse direction, turned to the right and to the left making full 360° turns. Upon completion of the turns the diameter of the circle made by the carrier shall be measured.

4.6.1.11 Hydraulic pressure at loader engine start-up test. To determine conformance to 3.5.18.1, the starter switch and engine shall be operated as specified and the hydraulic oil pressure gage checked for pressure requirements.

4.6.1.12 Loader electrical system check. To determine conformance to 3.5.18.2 through 3.5.18.9; the electrical system shall be operated and checked for functional requirements The system shall be checked for completeness or mounting, and for any other defect that may cause failure of the electrical systems

4.6.1.13 Controls checks. To determine conformance to 3.5. 19.1 through 3.5.19.5, the controls shall be operated as specified for specific controls and checked for functional requirements

4.6.1.14 Hydraulic system test. To determine conformance to 305.20*1 and 3.5.20.2$, the hydraulic. pump shall be operated for not less than 15 minutes and checked for operating pressures* Relief and proof pressures shall be checked using an external pressure source and gages. Burst pressure shall not be applied.

4.6.1.15 Superstructure operation (normal loader operation) test* To determine conformance to 3.5.21, loader superstructure shall be operated as specified and checked for functional requirements-

4.6.1.15*1 Missile release circuits tests To determine conformance to 3.5.21.1, the hoisting beams shall be placed in their latched positions, with no missile aboard, and a testing device similar to the one depicted on Figure 2 schematic, shall be plugged into the receptacle on the end of the index boom. The wiring harness which has been detached from the index boom, and the wiring harness which supplies main power to the instrument box, shall be plugged into the testing device and the following check shall be made:

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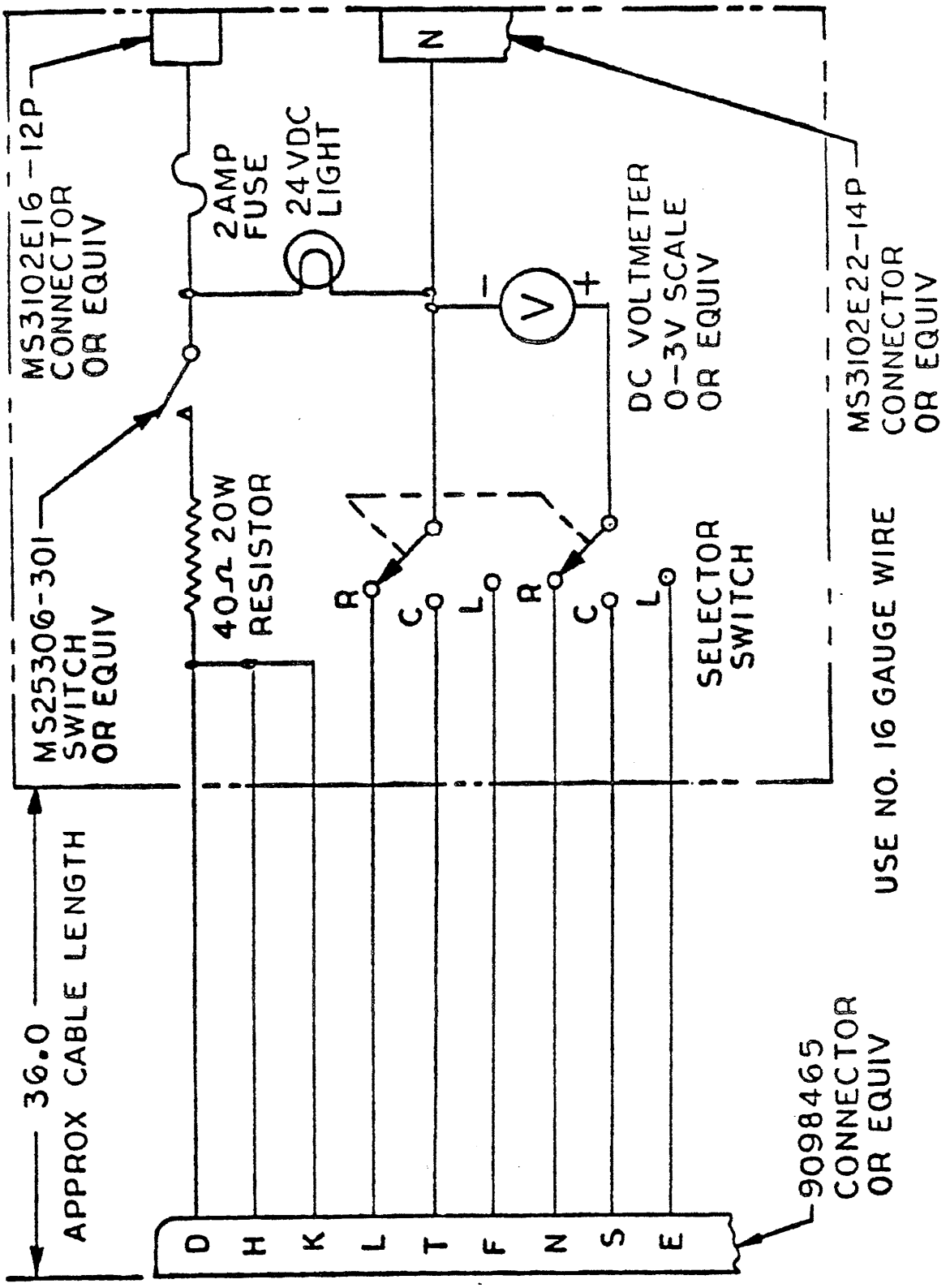


FIGURE 2. Missile release circuits test schematic.

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(a) Position the loader superstructure so that the index boom and transfer arm limit switches are closed (plungers extended) and place the master switch in the "ON" positions
(b) Place the Selector Switch on the "R" position* With the momentary switch on the test unit held in the closed position, the voltmeter shall read between 0.30 and 1.0 volt. A zero reading on the voltmeter indicates that the solenoid is not wired correctly.
(c) The test shall be repeated with the selector switch positioned on the remaining two positions.

4.6.1.16 superstructure operating speeds test. To determine conformance to 305.21.2, the superstructure shall be operated as specified and observed for functional requirements-

4.6.1.17 Superstructure. To determine conformance to 3.5.21.3, with the vehicle superstructure loaded, the elevation and extension cylinders positioned as specified, and the engine off, superstructure shall be checked for settling.

4.6.1.18 Azimuth and roll controls limitation tests. To determine conformance to 3.5.21.4 the azimuth and roll controls shall be operated and their effectiveness observed.

4.6.1.19 Reduced pressure control test. To determine conformance to (a) through (d) of 3.5.21.5 The superstructure shall be operated as specified and observed for functional requirements.

4.6.1.20 Manual latching of missiles test. To determine conformance to 3.5.21.6, the manual latching and locking devices shall be operated as specified and checked for positive locking.

4.6.1.21 Manual release of missiles test. To determine conformance to 3.5.21.7, with the latch-release solenoid manually depressed the missiles shall be checked to determine if fully released from the superstructures

4.6.1.22 Forward missile support test. To determine conformance to 3.5.21.8, the outboard forward missile supports shall be operated as specified and observed for functional requirements.

4.6.1.23 Rear center missile support test. To determine conformance to 3.5.21.9, the rear center missile support and safety bar shall be stowed and checked for loader shipment and crane operation.

4.6.1.24 Superstructure electrical system checks. To determine conformance to 3.5.22 through 3.5.22.5, with the main cable connected as specified, the circuits shall be operated as specified and checked for functional requirements.

4.6.1.25 Electrical release of missiles test. To determine conformance to 3.5.22.6 the missile latch-release shall be operated as specified and the electrical release checked for functional requirements-

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4.6.1.25.1 Latch release solenoid test. To determine conformance to 3.5.22.6.1, with the hoisting beam assemblies in specified condition, the release solenoids shall be energized, and the latch locks and latching assemblies observed for functional requirements.

4.6.1,26 Index boom and transfer arm limit switches test. To determine conformance to 3.5.22,7, the limit switches shall be operated as specified, and the solenoids and switches observed for functional requirements.

4.6.1.27 Loading and unloading missiles test. To determine conformance to 3.5.23 (b) and (c), the loader shall load and unload the launcher and pallet on specified slopes and be checked for functional requirements.

4.6.1.28 Loading and unloading pallet test. To determine conformancesto 3.5.24, the loader shall load and unload a pallet as specified and be checked for functional requirements.

4.6.1.29 Proof load test. To determine conformance to 3.5.24.1, the loader shall be subjected to the following proof load tests:

a. Suspension system shall be locked out prior to proof load testing and will remain in that position until all proof load testing is completed. Attach a 1000 pound (453.6 kg) ballast to the loader pintle hook prior to proof loading.

b. Increase the weight of each test missile to a minimum of 1526 pounds (692.2 kg) and a maximum of 1580 pounds (716.7 kg). The weights shall be added to maintain the missile center-of-gravity (cog.).

C* Index the loader to the pallet loaded with three missiles with weights and latch hoisting beam to missile, Install two (2) safety straps around hoisting beam and missiles. The straps shall be fastened to prevent the missiles from falling in event of failure, but shall not carry the missile weight during proof load testing.

- WARNING -

NOTE : Caution must be exercised during proof load testing by the operator to meter the control valves to prevent abrupt superstructure motions. Engine RPM to be 700 to 800 RPM.
CAUTION: Personnel not directly engaged in the proof loading exercises are not to enter an area of 30 feet (9.1 m) in any direction of the loader during proof load testing.
CAUTION: All proof load testing exercises will be conducted with the dri.verfs protective canopy in place and the operator under that canopy.

d. With the loader indexed to the pallet on the ground, retract the extension cylinder fully. Extend the extension cylinder until missiles seat on the pallet.

e. Repeat d above for two (2) cycles.

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f. With the loader indexed to the pallet on the ground, retract the extension cylinder fully and then break index by elevating the boom* Place missiles in the transport position.

CAUTION: Exercise caution to keep missile load as far aft on loader as possible by using elevation and extension controls in sequence.

g. Raise missiles enough to allow azimuth rotation of super-structure at 0 degrees. Move missiles to full azimuth right. Move missiles to full azimuth left.

h. Repeat g above for two (2) cyles and then center missiles in azimuth.

i. Index loader to pallets seat missiles on pallet, release latch on "C" missile only and remove straps from "Co' missile. Return superstructure with "A" and "B" missiles to transport position.

j. Position loader under a test fixture or crane and install sling to fixture or crane and loader superstructure yoke. Sling shall have enough slack to allow superstructure to roll fully right, but hold superstructure in the event of failure.

k. With missiles elevated sufficiently to permit roll motions, roll missiles fall right.

l. Repeat k above for two (2) cycles and then return to center.

m. Remove sling. Index loader to pallet and transfer missiles onto pallet. Remove straps, latch missiles to pallets release loader latches and retract loader superstructure.

n. Recheck loader alignment on pallet/alignment fixture.

4.6.2 Control tests (see 4.5.3).

4.6.2.1 Stowed material check. To determine conformance to 3.3.9, all manufacturers, depot, and troop on-vehicle-material (OVM) shall be stowed on a production vehicle. The OVM used shall be of the latest production available to the contractor, and the contractor shall be permitted to unpack and shall repack all OVM (requiring packaging) for domestic shipment only. The stowed OVM items shall be examined for installation and interference with the operation of the vehicle in any manner.

4.6.2.2 Missile transport test. To determine conformance to 3.3.11, the vehicle selected shall be subjected to a test of transporting specified number of missiles and checked for functional requirements.

4.6.2.3 Side slope test. To determine conformance to 3.5.10 the loader shall be operated on specified slopes and observed for functional requirements.

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4.6.3′ Inspection comparison tests (see 4.5.4).

4.6.3.1 Environmental tests. To determine conformance to 3.3.1, the loader properly serviced and equipped, shall be subjected to applicable tests under conditions specified in Purchase Description MPD-7005.

4.6.3.2 Fuel system functional test. To determine conformance to 3.3.2, the loader shall be operated throughout all speed and gear ranges over cross country terrain and on smooth, level, hard-surfaced roads, and loader observed for functional requirements. The system shall be checked for leaks before and after road test. After road test the fuel tank shall be examined for foreign matter.

4.6.3.3 Lifting and towing test. To determine conformance to 3.3.12, the vehicle selected shall be subjected to a test of lifting and towing specified loads and observed for functional requirements.

4e6.3.4 Brake lever force test. To determine conformance to 3.5.9.1~ the brake lever shall be applied under specified low temperature and checked for force required to operate brake lever.

4.6.3.5 Trench crossing test. To determine conformance to 3.5.11, the loader shall be operated across trenches of specified width and observed for functional requirements.

4.6.3.6 vertical Obstacle test. To determine conformance to 3.5.12, the loader shall be operated through holes of specified diameter and depth, and observed for functional requirements.

4.6.3.7 Blast hole test. To determine conformance to 3.5.13, the loader shall be operated over rough terrain as specified, and terrain as shown in Figure 1.

4.6.3.8 Rough terrain test. To determine conformance to 3.5.14, the loader shall be operated over rough terrain as specified, and terrain as shown in Figure 1.

4.6.3.9 Operation in snow test. To determine conformance to 3.5.15, the loader shall be operated in snow no deeper than 12″ and shall meet the performance requirements of this specification.

4.6.3.10 Fording test. To determine conformance to 3.5.16, the loader shall be operated in water as specified and observed for functional requirements.

4.6.3.11 Towing test. To determine conformance to 3.5.17, the loader shall be operated at the specified speed towing the specified load, and shall be observed for functional requirements.

4.6.3.12 Loading and unloading missiles test. To determine conformance to 3.5.23 (a), the loader shall load and unload the launcher and pallet on specified slopes and be checked for functional requirements.

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4.6.3.13 Crane operation test. To determine conformance to 3.5.25.1, the crane shall be operated as specified and checked for functional requirements.

4.6.3.13.1 Crane control tests. To determine conformance to 3.5.25.2, the controls shall be operated as specified and observed for functional requirements.

4.6.3.13.2 Roll lock out tests. To determine conformance to 3.5.25.3, with roll-lockout lever positioned as specified, the roll lockout shall be checked for positive lockout.

4.6.3.13.3 Hoisting cable check. To determine conformance to 3.5.25.4, the extension control lever shall be operated as specified, the cable shall be observed for functional requirements.

4.6.3.13.4 Boom swing test. To determine conformance to 3.5.25.5, the operation of the azimuth control lever shall be observed for the functional requirements of the boom.

4.6.3.13.5 Boom elevation test. To determine conformance to 3.5.2506, the operation of the elevation control lever shall be observed for the functional requirements of the boom.

4.6.3.13.6 Hoisting hook position check. To determine conformance to (a) and (b) 3.5.25.7, the hoisting hook shall be positioned as specified and checked for distance from the ground.

4.6.3.13.7 Rated capacity check. To determine conformance to 3.5.25.8, the loader used as a crane shall be operated as specified and observed for operational requirements {Caution: Use the crane sling to restrain the load, and do not lift load more than one foot above the ground.)

4.6.3.14 Radio interference suppression test. To determine conformance to 3.5.26, the vehicle shall be subjected to a radio interference suppression test.

4.6.4 Auxiliary equipment. The auxiliary equipment specified in 6.3, shall be used to check the comparability of the loader with this equipment.

5. PACKAGING

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with level A and B of MIL-L-45779 (see 6.2).

6. NOTES

6.1 Intended use. The loader-transporter is intended to transfer HAWK guided missiles from pallets to launcher and vice versa, and to otherwise handle missiles and pallets as may be necessary within a missile battery. It is also intended for limited use as a crane and for short distance towing.

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6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and data of this document.
b. If first article samples are riot required (see 3.1).
c. Stowing of On-Vehicle-Material if required (see 3.3.9).
d. Preproduction inspection, if required (see 4.4.1).
e. Initial production inspection, if not required (see 4.4.2).
f. Simulated missile payload if required (see 4.6).
G. Degree of preservation, packaging, and packing required (see 5.1).

6.3 Auxiliary equipment (see 4.6.4)

	Drawing
Guided Missile, Air Defense: XM3EI Standard Type	9071755
Launcher, Zero Length, Guided Missile:	9098880
Pallet, Loading and Storage, Guided Missile:	9196208
Truck, Cargo, 2-1/2 Ton, M36	8750400
Chassis, Trailer, 2 Ton, M390C	8736315
Ramp	9098330
Crane	9098300
Adapter, hoist	10916330

6.4 Definitions. For the purposes of this specification the following definitions shall apply:

a. Normal loader operation. Normal loader operation shall refer to operation associated with the loaders primary tactical' function within a missile battery, namely that of transferring 1, 2, and 3 missiles at a time from storage pallets to launchers and vice versa, and transferring pallets from ground to XM390 trailers and the reverse. For such operations indexing forks shall point toward front of loader, superstructure shall respond to hydraulic controls as specified in 3.5.21 and suprestructure electrical circuits operate in accordance with 3.5.22, Normal operation shall involve indexing, positioning of superstructure on pallets or launchers for loading and unloading of missiles, latching Of missiles to superstructure and release therefrom, transport of missiles latched to superstructure and latching to, transporting and unloading empty pallets on XM390 trailers or the reverse, .

b. Normal loader transport position. Normal loader transport position shall refer to the position of the superstructure during missile transport by the loader (in normal loader operation) wherein the latched missile(s) are held firmly in loader saddles with full weight of each missile carried by the applicable front and rear loader missile supports.

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c. Final extension phase. Final extension phase shall refer to operation of superstructure in extension and retraction (in normal loader operation) during period when reduced extension pressure feature is operative as specified in 3.5.21.5. In missile transfer operations, loader is indexed to launcher or pallet during the final phase of extension.

de Crane operation. Crane operation shall refer to operation of the loader when loader is rigged with items as specified (including ramp) and adjusted for crane operation. Loader shall be adapted as follows for crane operation.

1. Place "forward and rear center missile supports in stowed positions (see 3.5.21.8 and 3.5.21.9).
2. Lockout carrier suspension system (see 3.5.19.2).
3. Unpin azimuth cylinder rod boom support, rotate boom through 3s200 roils and repin to azimuth cylinder.
4. Place roll lockout in "crane" position (see 3.5.25.3').
5. Extend superstructure until transfer arms contact link assembly and pin together using links provided.
6. Assemble crane attachments to ramp. l?lace ends of ramp in crane support bracket and pin.
7. Place crane support bracket on hoisting support boom trunnion shaft. Using quick release pins, pin front stays to boom support assembly and pin fairlead sheave assembly to center hoisting beam housing.
8. Using loader power, lift ramp with crane cable and pin rear cable stays to loader index forks.

6.5 Inspection equipment calibration and inspection. In inspecting the supplier's inspection equipment, the Government inspector will determine that the supplier has available, and utilizes correctly, gaging, measurings and test equipment of the required accuracy and precision. Except as otherwise specified the supplier will have available a set of master gages~ standards and appropria&e instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the supplier and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

6.6 Process average. Sampling may be initiated if the process average value for the first twenty vehicles inspected is less than the A@ specified in the classification of defects for major and minor defects. Process average shall be computed as follows:

$$\text{Process Average} = \frac{\text{Number of Defects}}{\text{Number of Vehicles Inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified AQL.

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6.7 The following shall be used with the classification of defects for leaks:

(a) Weep: Any evidence of fluid beyond the seal.
(b) Seep: Any evidence of fluid beyond the seal that does not result in formation of a droplet-
(c) Droplet: Any evidence of fluid beyond the seal that results in the formation of a droplet*
(d) Drip: Any evidence of fluid beyond the seal where droplet forms and drips.

6.8 Definitions of recurring major and minor defectse The Government Inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any other departures from good workmanship- The Government Inspector will assure that all deficiencies encountered during the inspection are enumerated on the Deficiency Sheet for the vehicle. The defects noted on Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specification and definitions contained in MIL-STD-105. Corrective action shall be taken for recurring deficiencies.

Custodian:
Army - AT

Review activity:
Army - Ml

Preparing activity:
Army - AT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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